Professional Git Essentials Guide

What is Git?

Git is a distributed version control system that tracks changes, enables collaboration, and preserves project history.

Core Concepts

Repository, Working Directory, Staging Area, Commit, Branch, Merge, Remote Repository.

Essential Commands

`git init`, `git clone`, `git status`, `git add`, `git commit -m`, `git log`, `git branch`, `git switch`, `git checkout`, `git merge`, `git pull`, `git push`, `git fetch`, `git diff`, `git restore`.

Typical Workflow

Clone → Edit → git status → git add → git commit → git pull → git push.

Best Practices

Commit often, use meaningful commit messages, work with branches, review changes before committing, and never commit sensitive data.

Conclusion

Git is an essential tool for modern software development and collaborative projects.